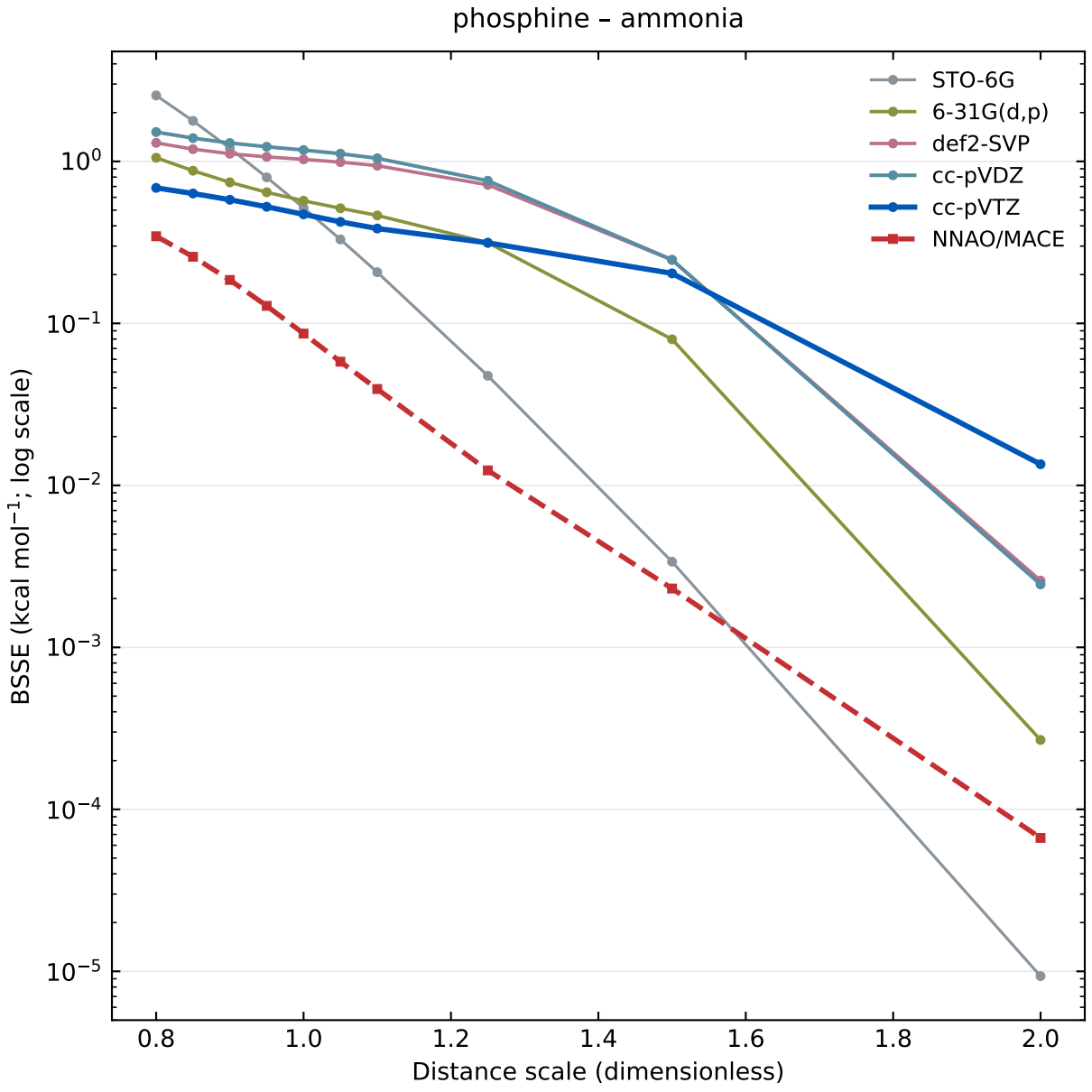
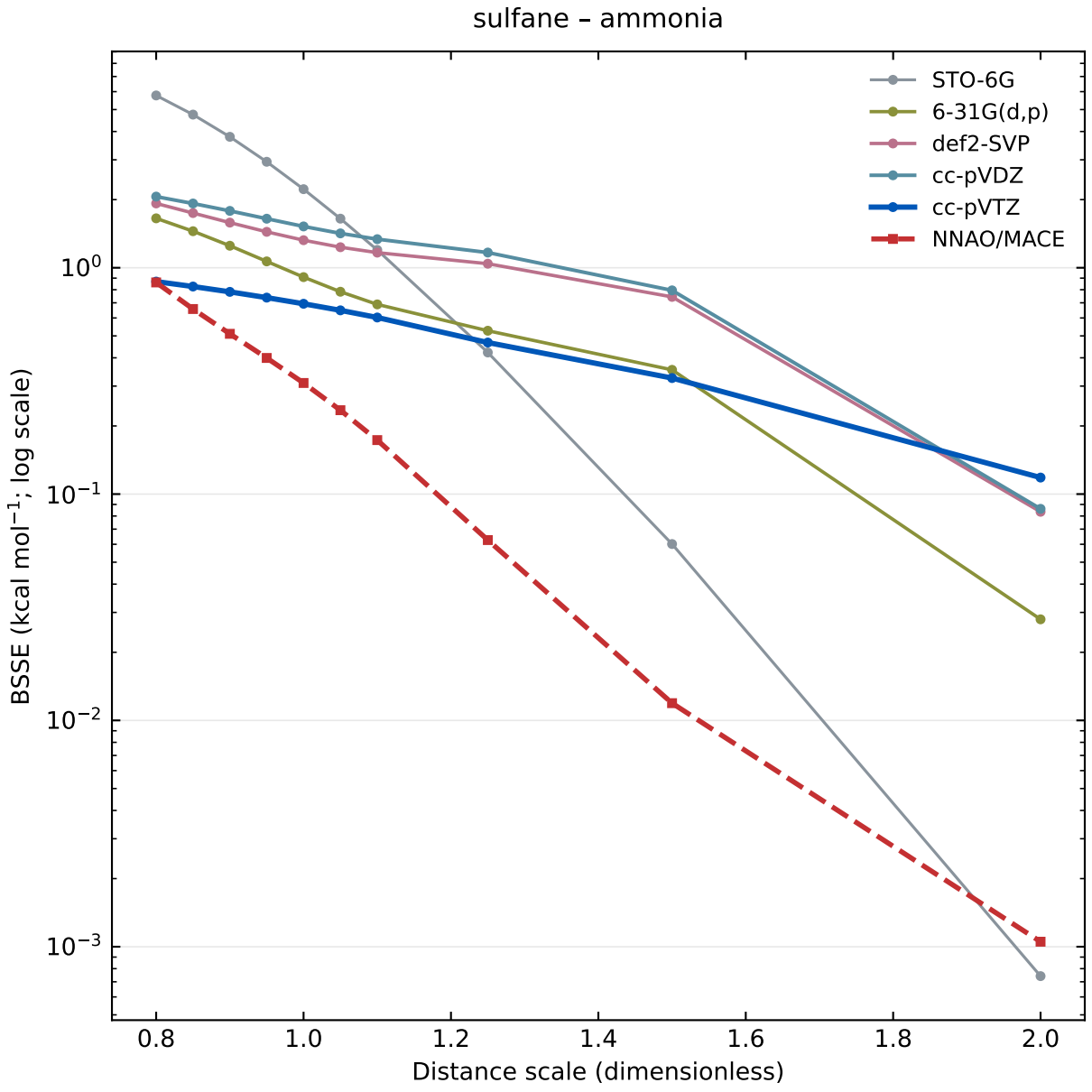
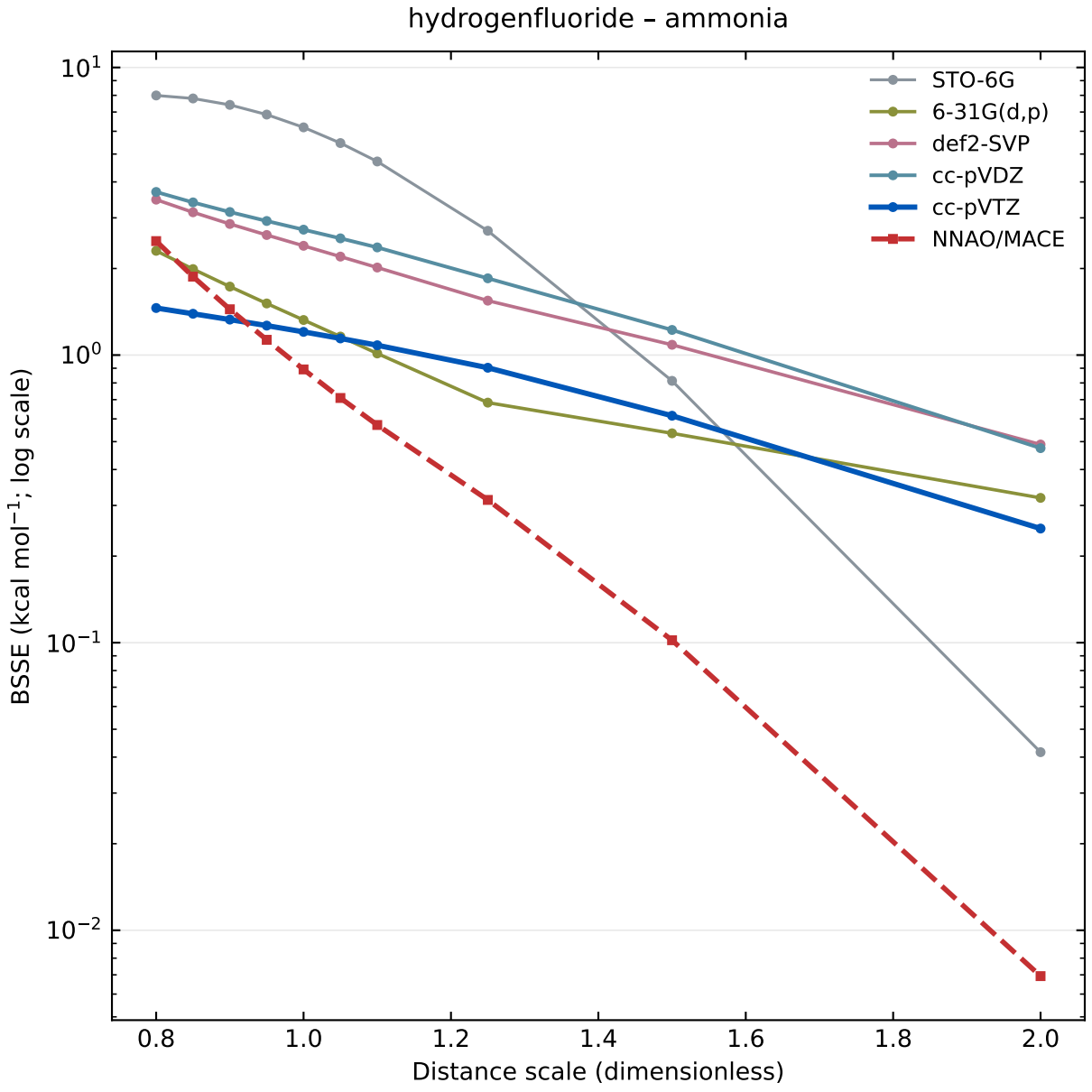
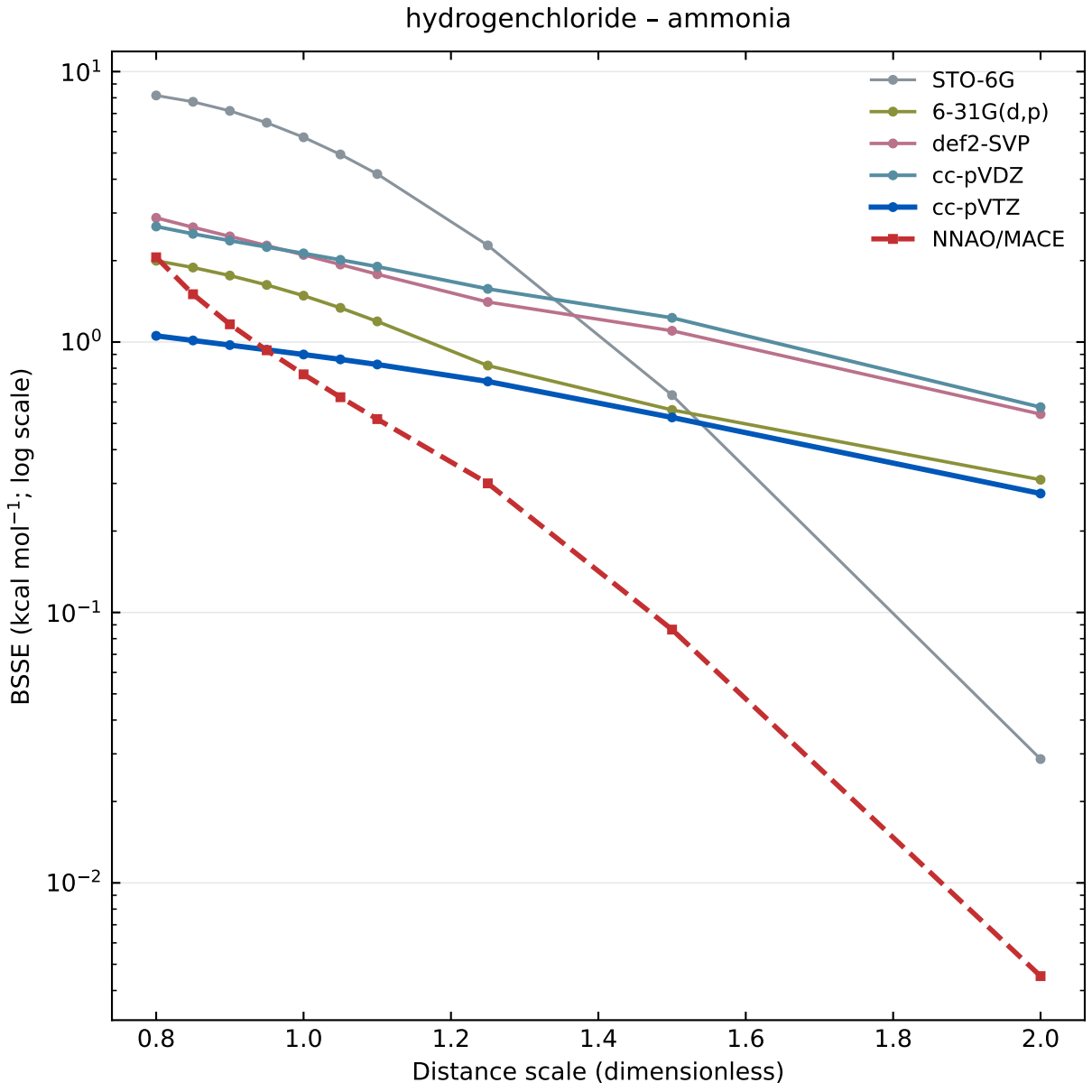
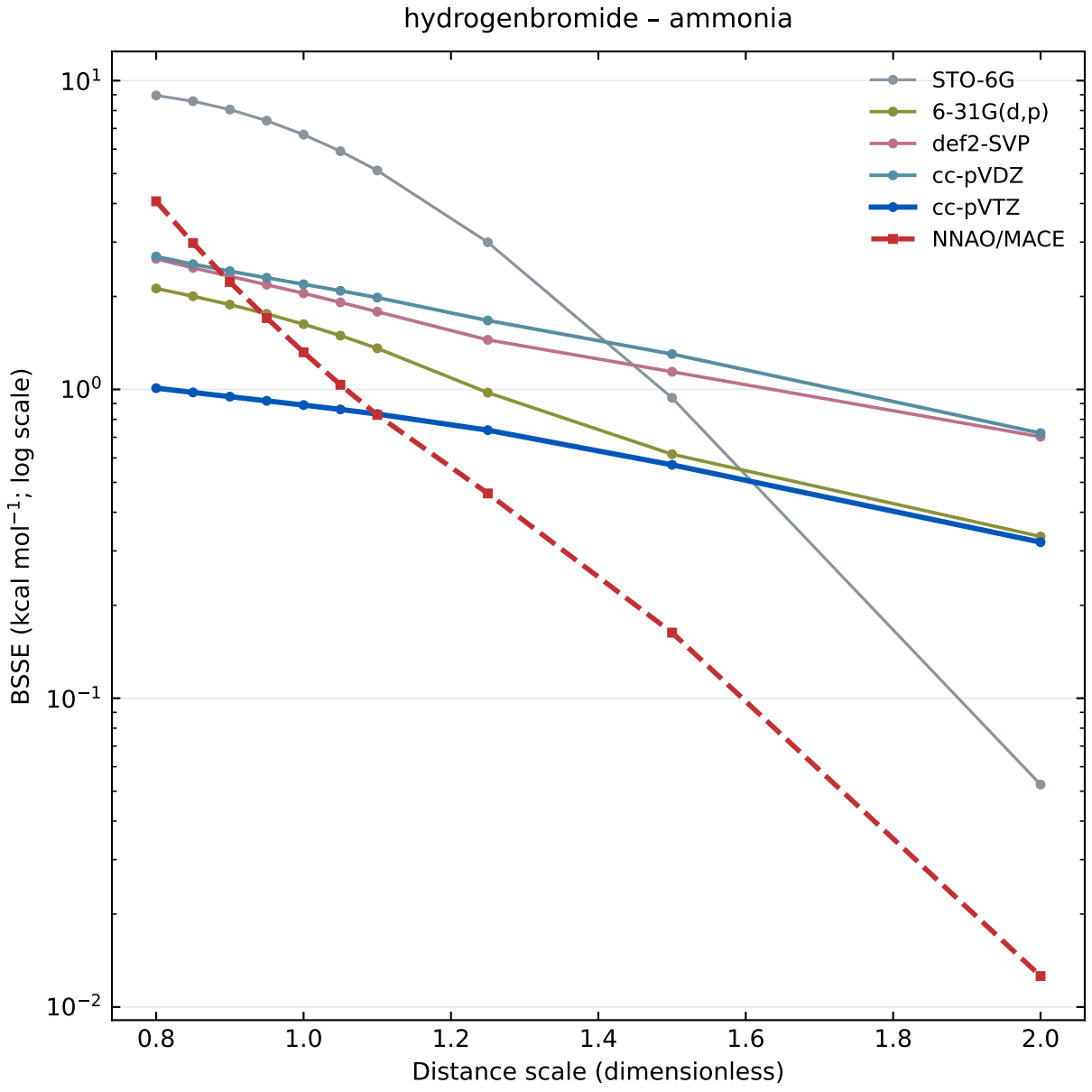
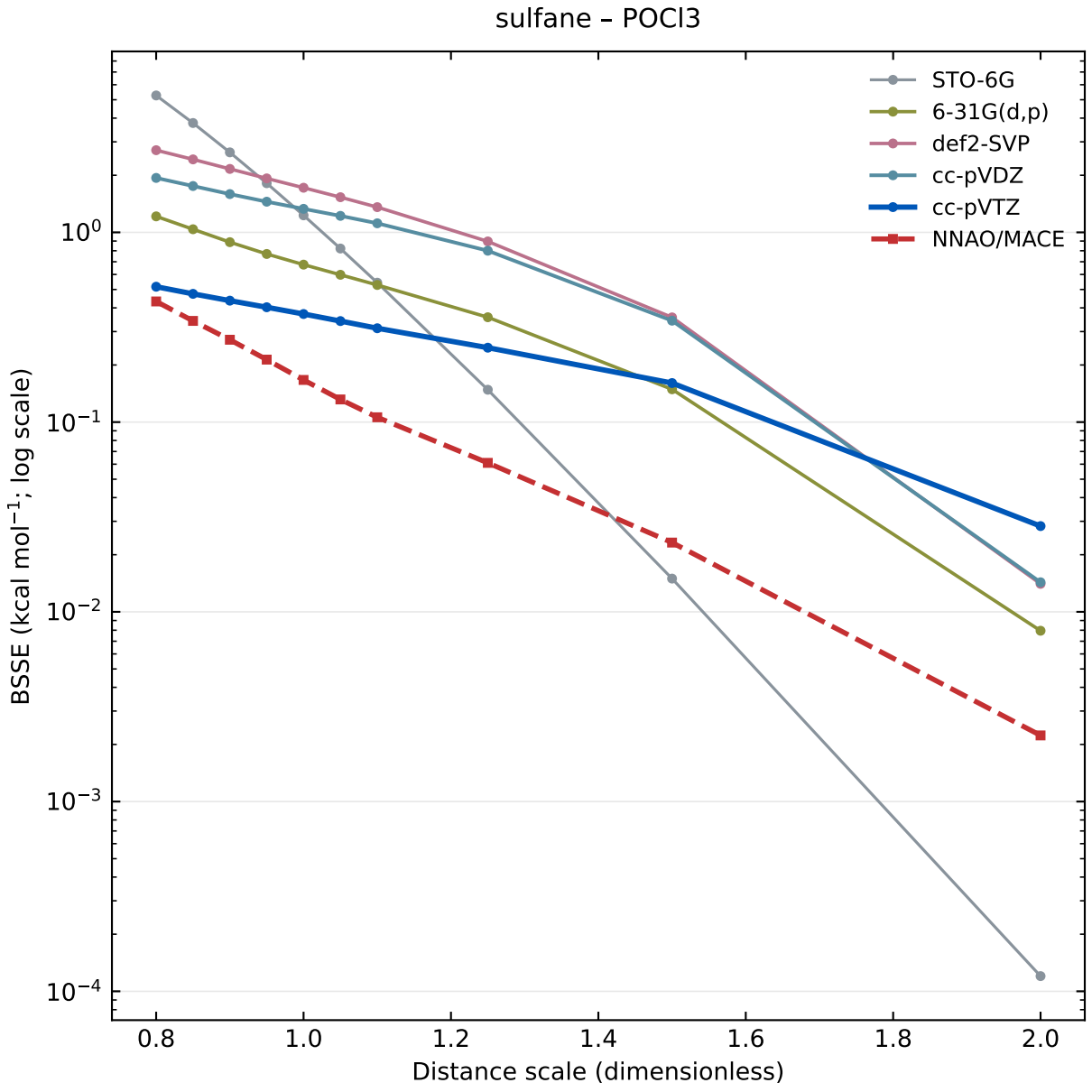

phosphine – ammonia
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol$^{-1}$; log scale)
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)

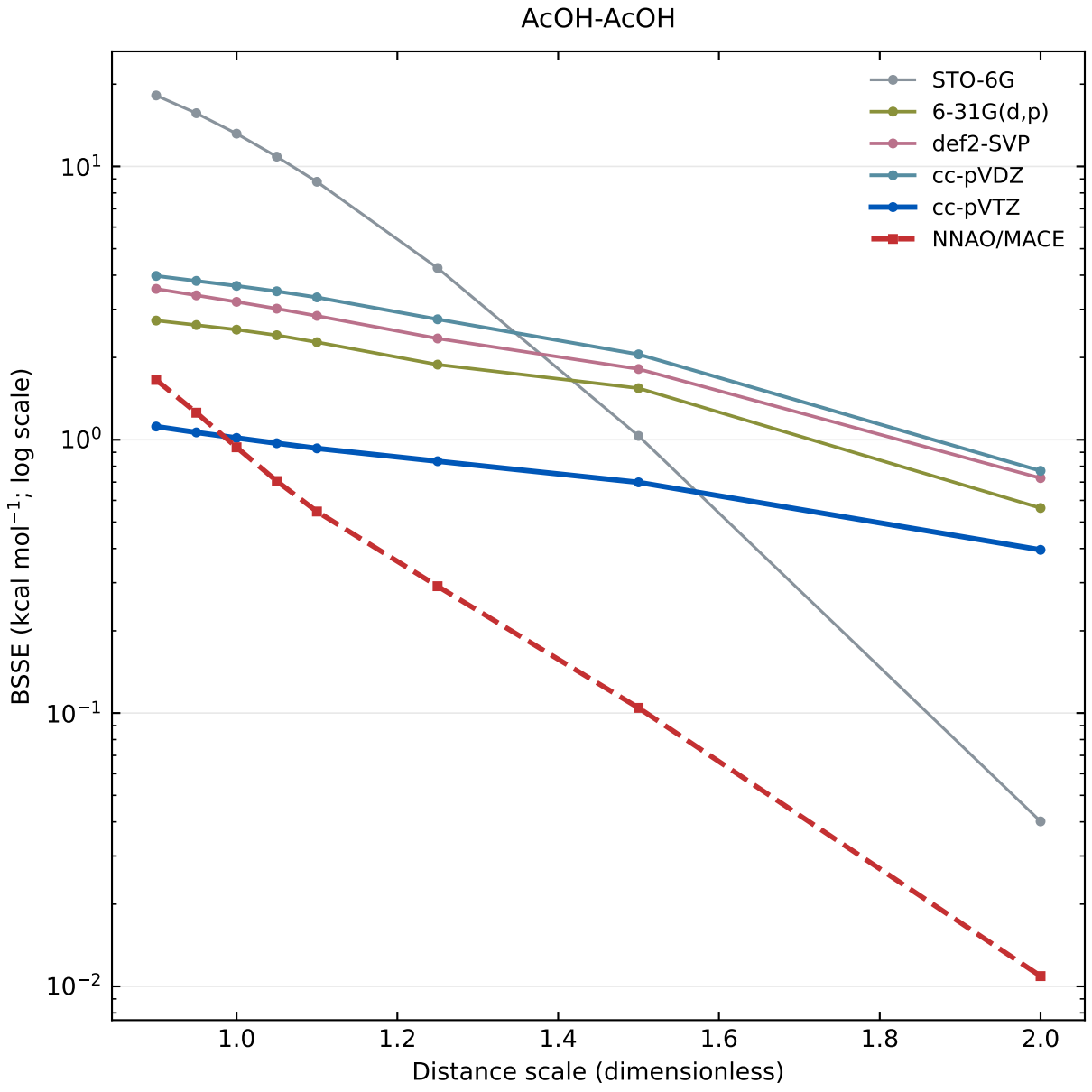

sulfane – ammonia
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol$^{-1}$; log scale)
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)

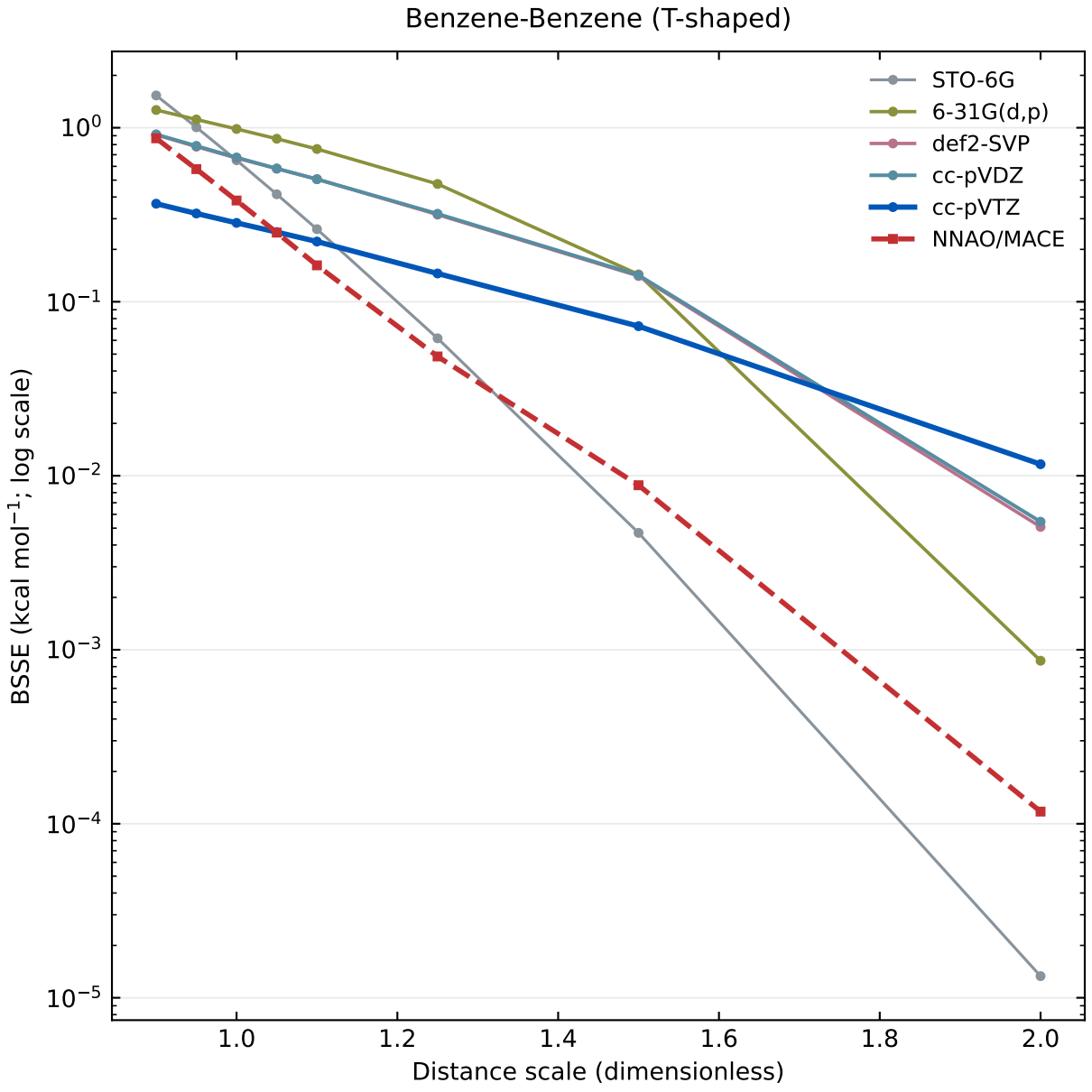

hydrogenfluoride – ammonia
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
10¹
10⁰
10⁻¹
10⁻²
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)

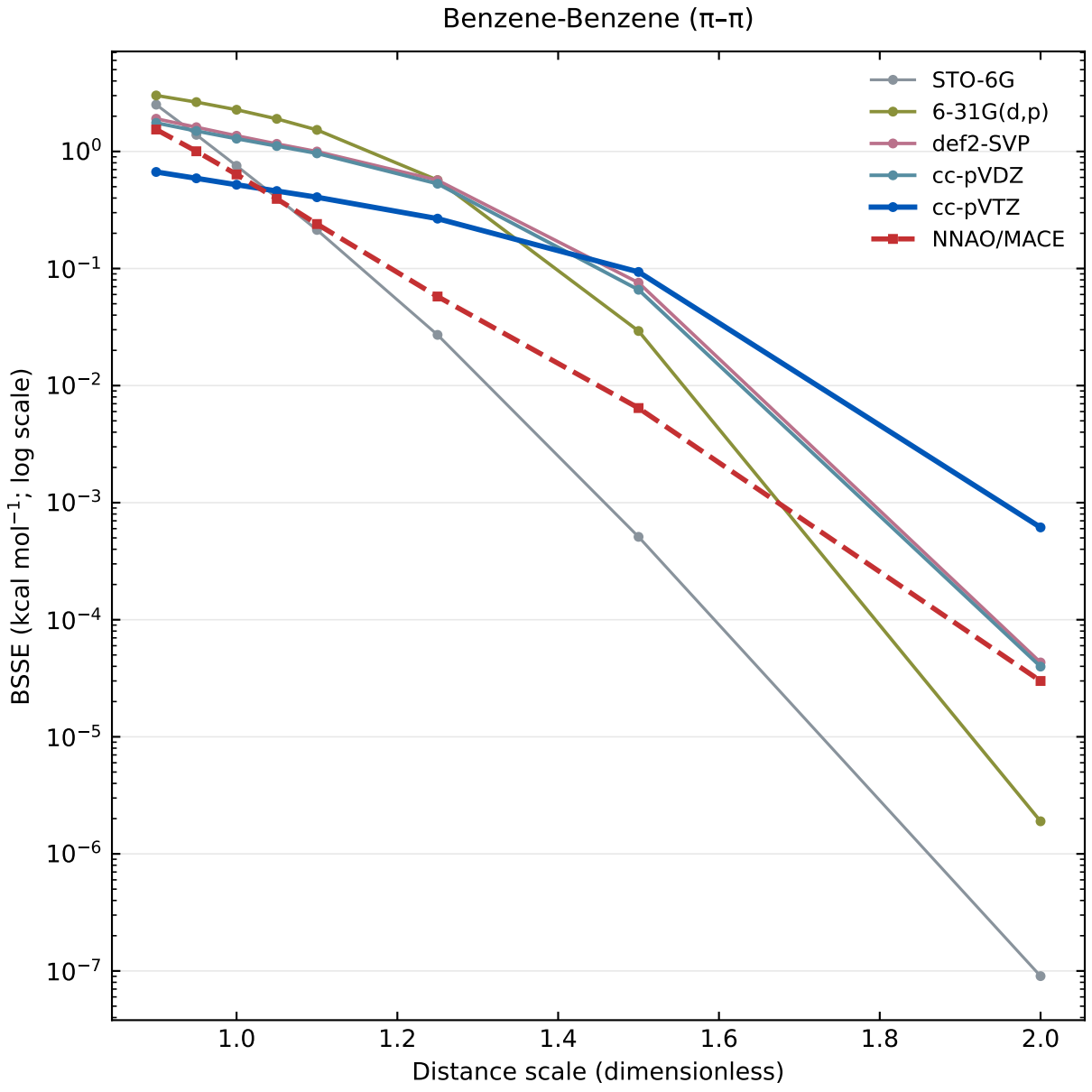

hydrogenchloride – ammonia
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
Distance scale (dimensionless)
10¹
10⁰
10⁻¹
10⁻²
0.8
1.0
1.2
1.4
1.6
1.8
2.0

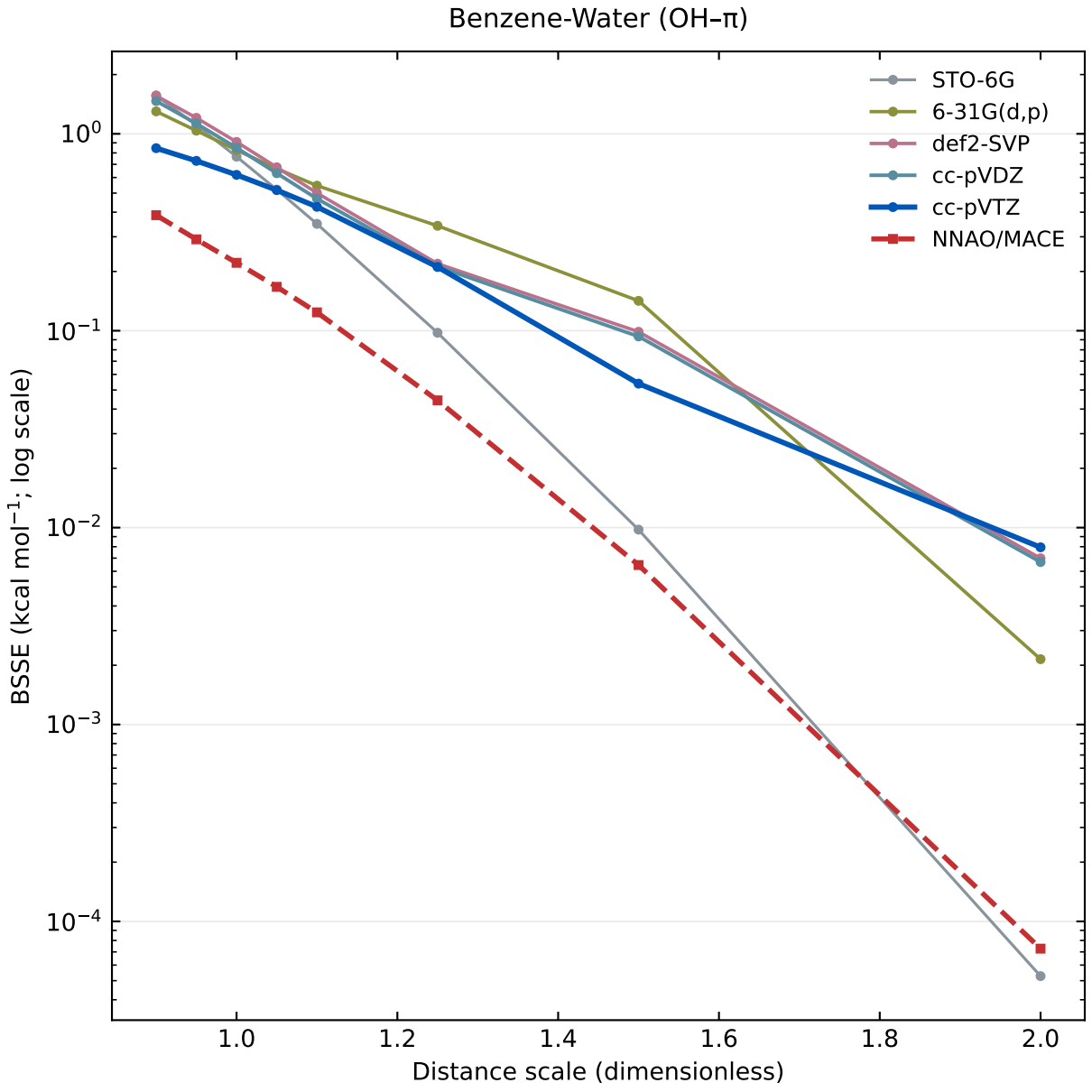

hydrogenbromide – ammonia
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
Distance scale (dimensionless)
10¹
10⁰
10⁻¹
10⁻²
0.8
1.0
1.2
1.4
1.6
1.8
2.0

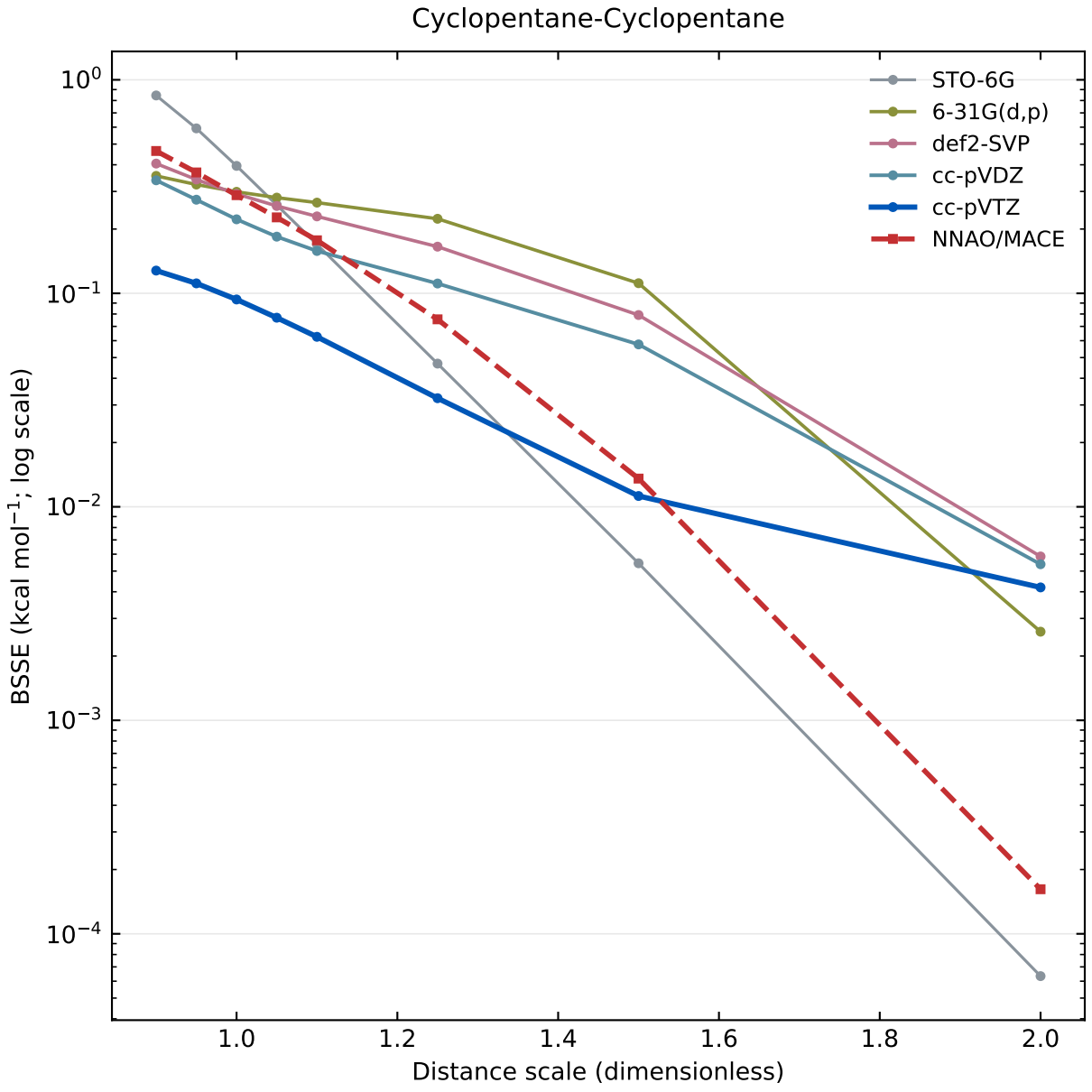

sulfane – POCl3
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)

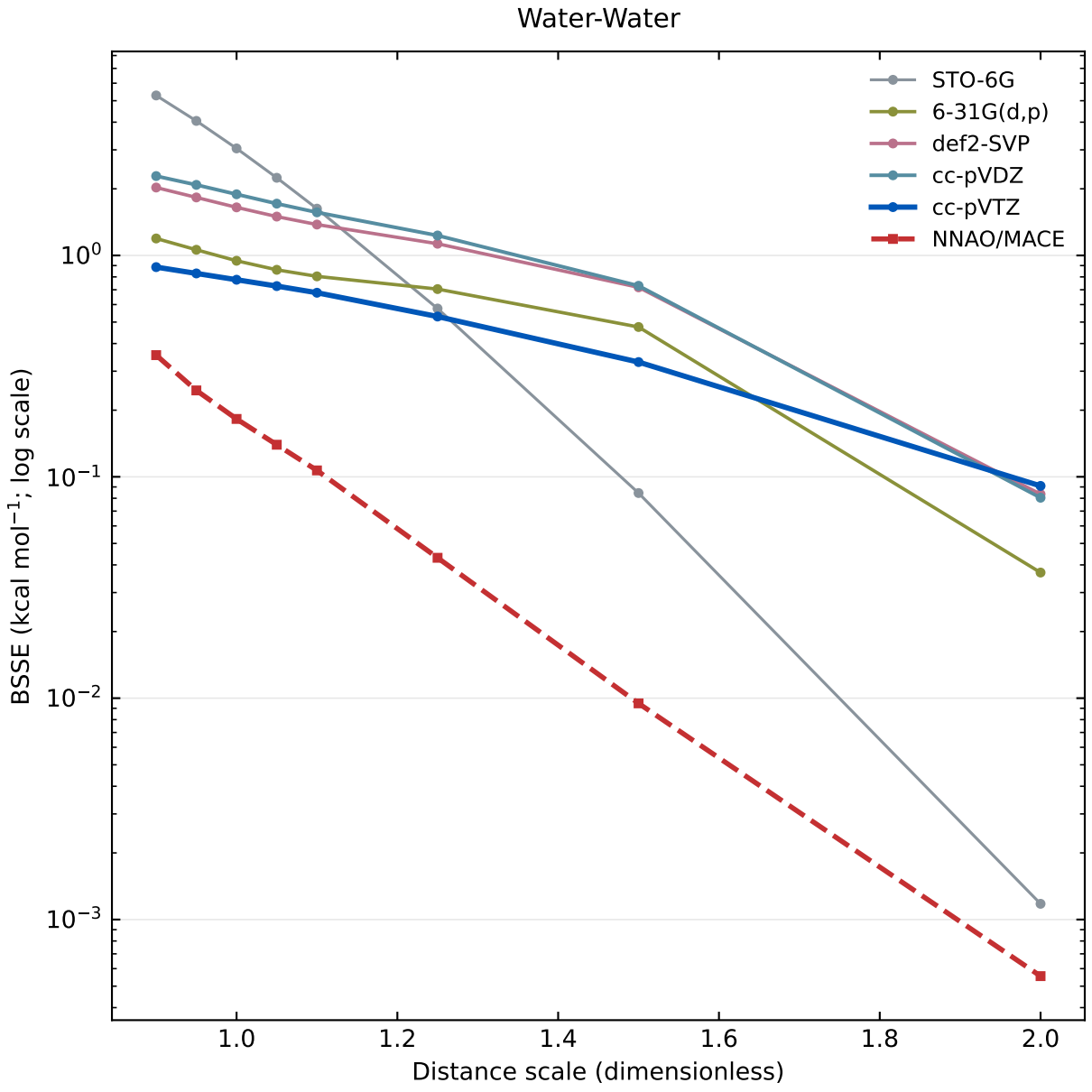

AcOH-AcOH
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol$^{-1}$; log scale)
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)


Benzene-Benzene (T-shaped)
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
Distance scale (dimensionless)
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
1.0
1.2
1.4
1.6
1.8
2.0


Benzene-Benzene (π–π)
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
10⁻⁶
10⁻⁷
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)


Benzene-Water (OH–π)
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
Distance scale (dimensionless)
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
1.0
1.2
1.4
1.6
1.8
2.0


Cyclopentane-Cyclopentane
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol⁻¹; log scale)
Distance scale (dimensionless)
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
1.0
1.2
1.4
1.6
1.8
2.0


Water-Water
STO-6G
6-31G(d,p)
def2-SVP
cc-pVDZ
cc-pVTZ
NNAO/MACE
BSSE (kcal mol$^{-1}$; log scale)
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
1.0
1.2
1.4
1.6
1.8
2.0
Distance scale (dimensionless)